

AI for Bharat Hackathon

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Team Name : Team Meridian

Team Leader Name : Gayatri Srinivasan

Problem Statement : Disconnected supply chains in Indian retail markets lead to significant inventory waste and missed revenue opportunities due to a lack of synchronization between local inventory and global/hyper-local market signals.

Brief

The Meridian Commerce Engine is an AI-powered anticipatory commerce system designed specifically for "Bharat."

Core Function: It transforms traditional inventory management into a predictive model by synchronizing local stock with global trends and hyper-local variables (weather, festivals, transit).

Key Innovation: Combines advanced **Temporal Fusion Transformers (TFT)** for demand forecasting with **Semantic Sentiment Analysis** to turn unstructured social and economic data into actionable retail intelligence.

Solution Impact & USP

How it is different: Unlike generic inventory tools, Meridian features a "Trend Bridge" and "Local Affinity Filter" that adjust global signals based on regional cultural preferences and demographics unique to Indian districts.

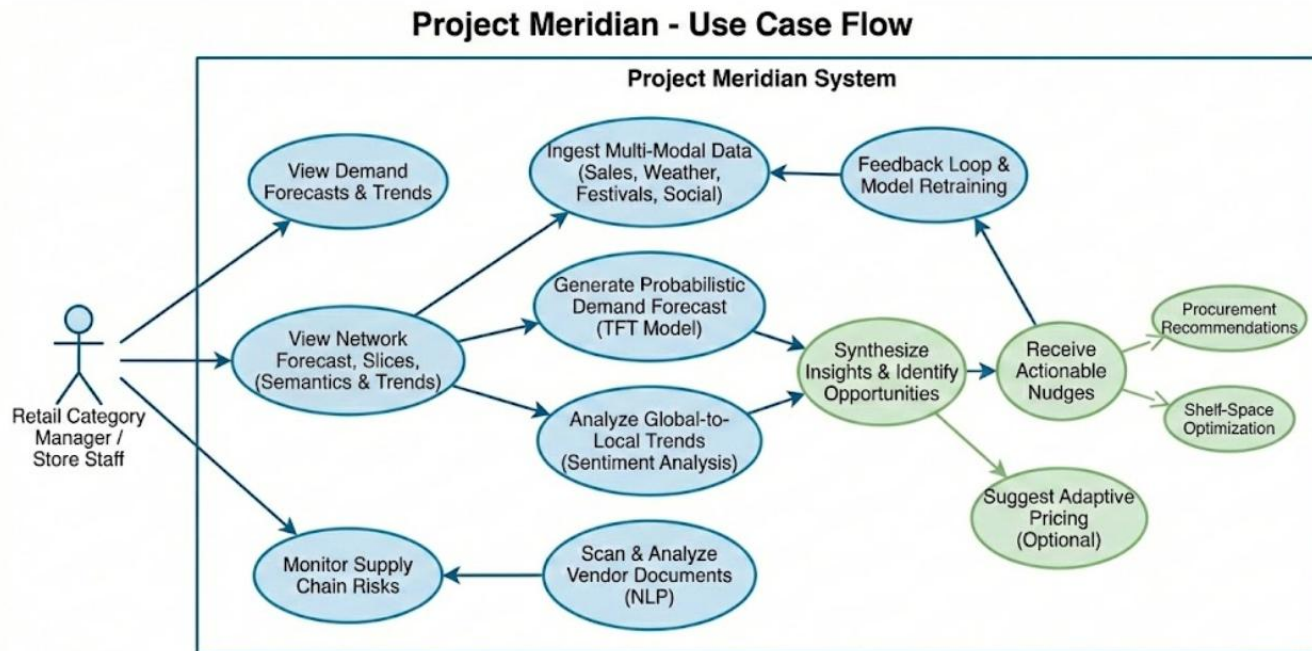
Problem Solving: It prevents stockouts and overstocking by providing probabilistic forecasts with 95% confidence intervals and automating procurement before demand spikes occur.

USP: A unified platform that integrates multi-modal data (social media, news, environmental feeds) with a natural language Virtual Category Manager copilot, making high-end AI accessible to retailers without technical expertise.

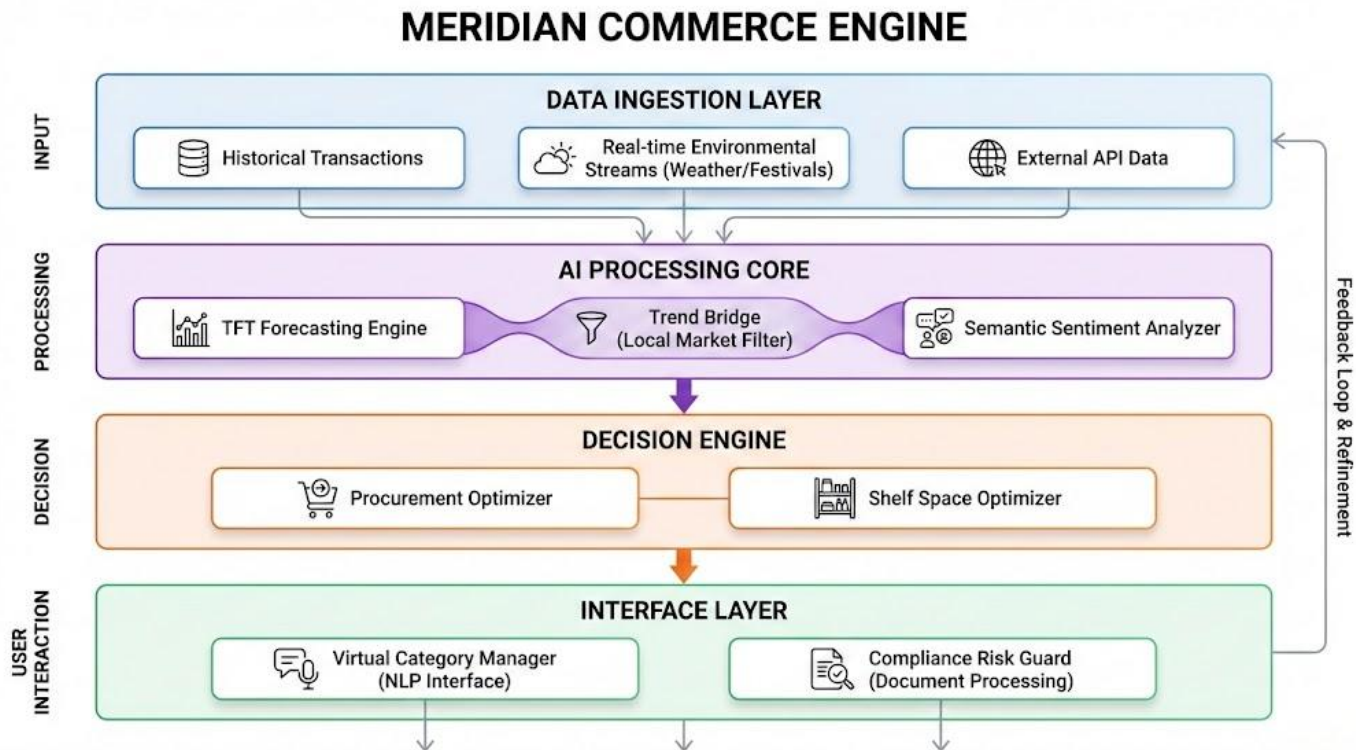
Features

- ✓ **High-Precision Forecasting:** Probabilistic demand predictions using TFT Core with best/worst/most-likely scenarios.
- ✓ **Global-to-Local Trend Analysis:** Extraction of trend vectors from social media and macro-economic data.
- ✓ **Virtual Category Manager:** A natural language interface (Copilot) for inventory queries and automated "Action Nudges."
- ✓ **Automated Procurement:** Auto-generation of purchase orders based on 14-day predicted spikes, considering supplier reliability and lead times.
- ✓ **Compliance Risk Guard:** Automated processing of invoices and logistics documents (PDF/Images) with anomaly detection.
- ✓ **Self-Correcting Loop:** A closed-loop system that automatically adjusts model weights based on actual sales performance.
- ✓ **Bharat-Ready Tech:** Support for regional languages/currencies and a robust offline mode for limited connectivity environments

Use Case Diagram



Architecture Diagram



Technology:

AI/ML: Temporal Fusion Transformers (TFT) for time-series forecasting; Multi-modal Sentiment Analysis (NLP).

Languages & Frameworks: Python-based implementation; Hypothesis for property-based testing.

Data Storage: Time Series DB (transactions), Graph DB (supplier networks), and Document Stores (social media feeds).

Infrastructure: Microservices architecture; Automated horizontal scaling for diverse retail volumes.

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Thank You

